Untitled44.ipynb

[18]

```python
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from google.colab import files
import io
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression, Ridge, Lasso
from sklearn.preprocessing import PolynomialFeatures, StandardScaler
from sklearn.metrics import mean_squared_error, r2_score

uploaded = files.upload()
file_name = list(uploaded.keys())[0]
data = pd.read_csv(io.BytesIO(uploaded[file_name]))
data = data.dropna()

X = data.drop("medv", axis=1)
y = data["medv"]

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

linear_model = LinearRegression()
linear_model.fit(X_train, y_train)
y_pred_linear = linear_model.predict(X_test)
```

Choose files BostonHousing.csv

BostonHousing.csv(text/csv) - 35177 bytes, last modified: 28/02/2026 - 100% done

Saving BostonHousing.csv to BostonHousing (3).csv

Untitled44.ipynb

[19]

```python
poly = PolynomialFeatures(degree=2)
X_poly_train = poly.fit_transform(X_train)
X_poly_test = poly.transform(X_test)
poly_model = LinearRegression()
poly_model.fit(X_poly_train, y_train)
y_pred_poly = poly_model.predict(X_poly_test)

ridge_model = Ridge(alpha=1.0)
ridge_model.fit(X_train_scaled, y_train)
y_pred_ridge = ridge_model.predict(X_test_scaled)

lasso_model = Lasso(alpha=0.1)
lasso_model.fit(X_train_scaled, y_train)
y_pred_lasso = lasso_model.predict(X_test_scaled)

def evaluate(name, y_true, y_pred):
    print(name)
    print("MSE:", mean_squared_error(y_true, y_pred))
    print("R2 Score:", r2_score(y_true, y_pred))
    print()

evaluate("Linear Regression", y_test, y_pred_linear)
evaluate("Polynomial Regression", y_test, y_pred_poly)
evaluate("Ridge Regression", y_test, y_pred_ridge)
evaluate("Lasso Regression", y_test, y_pred_lasso)

models = {
    "Linear": y_pred_linear,
    "Polynomial": y_pred_poly,
    "Ridge": y_pred_ridge,
    "Lasso": y_pred_lasso
}
```

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```python
models = {
    "Linear": y_pred_linear,
    "Polynomial": y_pred_poly,
    "Ridge": y_pred_ridge,
    "Lasso": y_pred_lasso
}
```

```
Linear Regression
MSE: 20.687720473048476
R2 Score: 0.7200277678580317

Polynomial Regression
MSE: 12.458958339477283
R2 Score: 0.831389718310843

Ridge Regression
MSE: 20.65820201533079
R2 Score: 0.7204272487243444

Lasso Regression
MSE: 21.63888776463179
R2 Score: 0.7071553767160526
```

```python
for name, preds in models.items():
    plt.figure()
    plt.scatter(y_test, preds)
    plt.plot([y_test.min(), y_test.max()], [y_test.min(), y_test.max()])
    plt.xlabel("Actual MEDV")
    plt.ylabel("Predicted MEDV")
    plt.title(name)
    plt.show()
```

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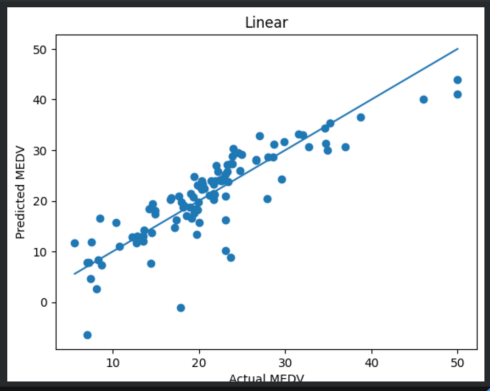
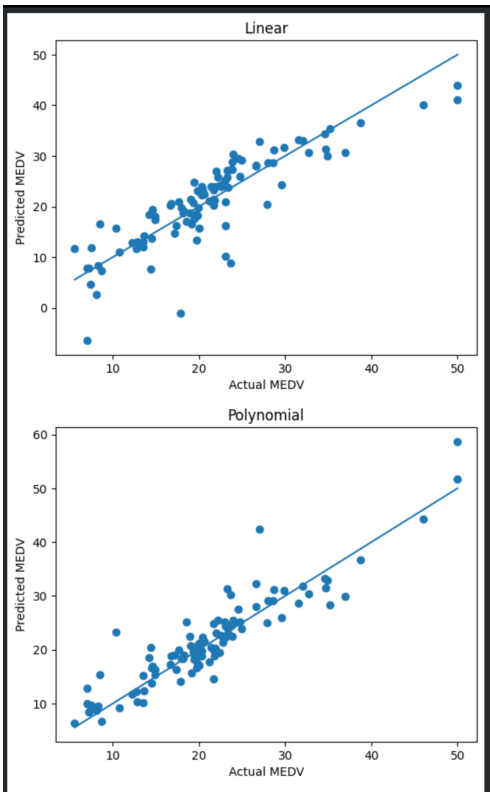
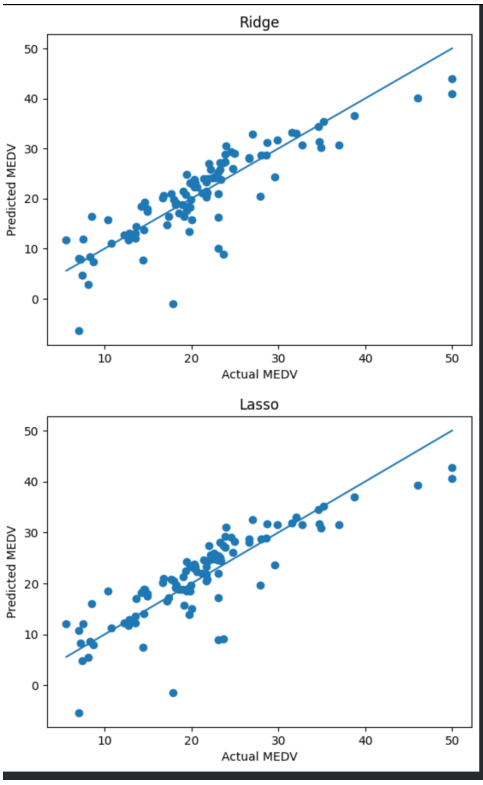
```




Linear
Predicted MEDV
Actual MEDV
Polynomial
Predicted MEDV
Actual MEDV


Ridge
Predicted MEDV
Actual MEDV
Lasso
Predicted MEDV
Actual MEDV